

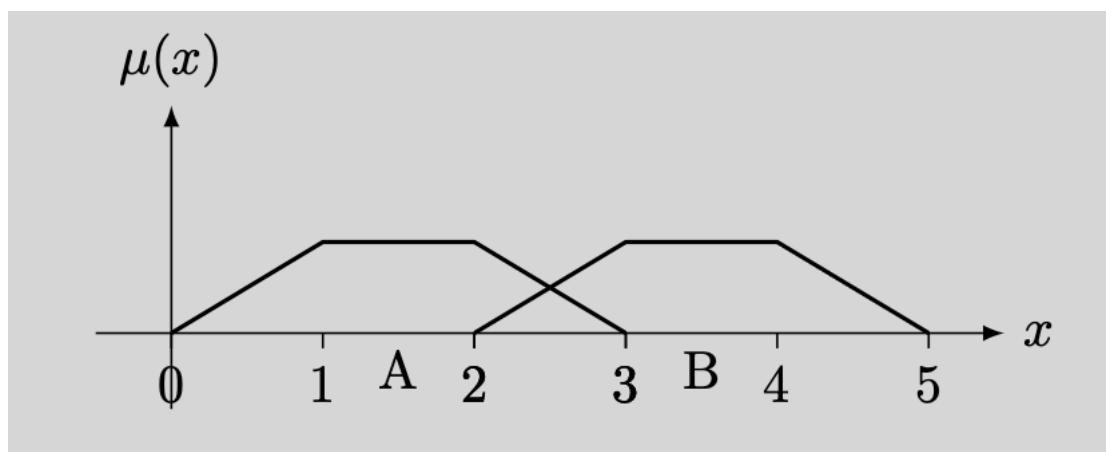
Fuzzy Systems and Evolutionary Computing 2022/23

Open Questions

- What is a linguistic variable? Give an example.
- What are fuzzification and defuzzification in a fuzzy rule based system? Which is the connection with a frame of cognition (aka, fuzzy partition)?
- Explain how fuzzy c-means works. If helpful, you can consider the two centroids $c_1 = (0, 1)$, $c_2 = (1, 0)$, $w = 2$ and three objects $x_1 = (0.75, 0.5)$, $x_2 = (0, 0)$ and $x_3 = (0.3, 1.1)$.
- Define n-point crossover. Give an example.
- Mutation in EA: what is it? Which is its role? Give an example with integer representation
- Give a high-level (=no need of mathematical details) explanation of Particle Swarm Optimization

Exercises

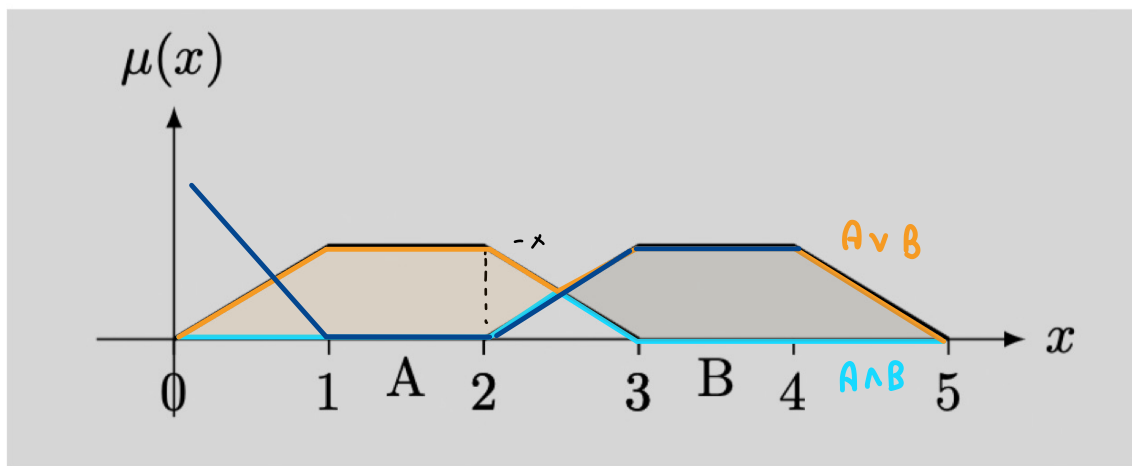
- 1) Let us consider the two fuzzy sets A and B given in the following figure



- Give the equation of the two functions defining A and B
 - Compute the intersection and union of f and g by means of a t-norm/t-conorm at your choice
 - Compute the complement of A
 - Compute support and core of G
- 2) Let us consider the fuzzy set g defined as $g(5)=0.3$, $g(6)=0.5$, $g(7)=0.9$, $g(8)=1$, $g(9)=0.9$, $g(10)=0.5$, $g(11)=0.3$ $g(x) = 0$ for all other values of x .
- Compute core and support of g
 - Compute the complement of g
 - Compute the cardinality of g
 - Compute the α -cut of g with $\alpha=0.6$
 - Compute the entropy of g
 - Compute the measure of fuzziness of g using the Euclidean distance
 - Compute the ordered weighted average with $w_1 = w_2 = w_3 = 2/14$, $w_4 = w_5 = 1/14$, $w_6 = w_7 = 3/14$
 - Compute the center of gravity of g

- 3) Given two propositions P and Q, with truth value $v(P)=0.7$ and $v(Q)=0.4$, compute
 - a) The logical *and* using product t-norm
 - b) The logical *or* using product t-conorm
 - c) $P \rightarrow Q$, using an implication at your choice
- 4) Let us consider a population of integer expressions (for instance $(3+4)-(6+8)$) with operations in $\{+, -, *, /\}$ and numbers in $\{0, \dots, 9\}$. Apply simple Genetic Programming with the following parameters
 - a) Random initial population of 4 individuals generated with full method and depth = 3
 - b) The fitness function is defined as $f(x) = x + 2$ with x the result of the expression represented by the tree
 - c) Generate 4 new individuals: two are obtained by crossover of the two most fit parents and two by mutation of the other two parents
 - d) Survivor selection: generational replacement
- 5) Let us consider a population of individuals represented by 6 binary digits. Apply simple Genetic Programming with the following parameters:
 - a) Generate random initial population of 4 individuals
 - b) Generate 4 new individuals: parents selected by roulette wheel selection, 2-point crossover and bit-wise mutation with $p_m = 1/6$
 - c) Survivor selection: age based
 - d) Create a new offspring with tournament selection and $k=2$

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a)

$$A: \begin{cases} x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x \leq 2 \\ -x+3 & \text{if } 2 \leq x \leq 3 \\ 0 & \text{OTHERWISE} \end{cases} \quad B: \begin{cases} x-2 & \text{if } 2 \leq x \leq 3 \\ 1 & \text{if } 3 \leq x \leq 4 \\ x-5 & \text{if } 4 \leq x \leq 5 \\ 0 & \text{OTHERWISE} \end{cases}$$

b) INTERSECTION $f \wedge g$ MIN

$$\bullet (A \cap B)(x) = \begin{cases} x-2 & \text{if } 2 \leq x \leq 2.5 \\ 3-x & \text{if } 2.5 \leq x \leq 3 \\ 0 & \text{OTHERWISE} \end{cases}$$

UNION $f \vee g$ MAX

$$\bullet (A \cup B)(x) = \begin{cases} x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x \leq 2 \text{ OR } 3 \leq x \leq 4 \\ -x+3 & \text{if } 2 \leq x \leq 2.5 \\ x-2 & \text{if } 2.5 \leq x \leq 3 \\ 5-x & \text{if } 4 \leq x \leq 5 \\ 0 & \text{OTHERWISE} \end{cases}$$

COMPLEMENT

$$\bullet A^c = \begin{cases} 1-x & \text{if } 0 \leq x < 1 \\ 0 & \text{if } 1 \leq x \leq 2 \\ x-2 & \text{if } 2 \leq x \leq 3 \\ 1 & \text{OTHERWISE} \end{cases}$$

SUPPORT of B

$$\{x \in B \mid B(x) > 0\} = \{2, 5\}$$

CORE of B

$$\{x \in B \mid B(x) = 1\} = \{3, 4\}$$

- 2) Let us consider the fuzzy set g defined as $g(5)=0.3$, $g(6)=0.5$, $g(7)=0.9$, $g(8)=1$, $g(9)=0.9$, $g(10)=0.5$, $g(11)=0.3$, $g(x) = 0$ for all other values of x .
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$$g(5) = 0.3$$

$$g(6) = 0.5$$

$$g(7) = 0.9$$

$$g(8) = 1$$

$$g(9) = 0.9$$

$$g(10) = 0.5$$

$$g(11) = 0.3$$

$$g(x) = 0 \text{ OTHERWISE}$$

CORE of g

$$\text{SUPPORT of } g = \{5, 6, 7, 8, 9, 10, 11\}$$

CARDINALITY of g

$$0.3 + 0.5 + 0.9 + 1 + 0.9 + 0.5 + 0.3 = 4.4$$

α -CUT WITH $\alpha = 0.6$

$$\{7, 8, 9\}$$

COMPLEMENT of g

$$g^c(5) = 0.7$$

$$g^c(6) = 0.5$$

$$g^c(7) = 0.1$$

$$g^c(8) = 0$$

$$g^c(9) = 0.1$$

$$g^c(10) = 0.5$$

$$g^c(11) = 0.7$$

$$g^c(x) = 1 \text{ OTHERWISE}$$

ENTROPY

$$H = -0.3 \cdot \log\left(\frac{1}{0.3}\right) - 0.5 \cdot \log\left(\frac{1}{0.5}\right) - 0.9 \cdot \log\left(\frac{1}{0.9}\right) - 1 \cdot \log\left(\frac{1}{1}\right) - 0.9 \cdot \log\left(\frac{1}{0.9}\right) - 0.5 \cdot \log\left(\frac{1}{0.5}\right) - 0.3 \cdot \log\left(\frac{1}{0.3}\right) \\ - (1-0.3) \cdot \log\left(\frac{1}{1-0.3}\right) - (1-0.5) \cdot \log\left(\frac{1}{1-0.5}\right) - (1-0.9) \cdot \log\left(\frac{1}{1-0.9}\right) - (1-0) \cdot \log\left(\frac{1}{1-0}\right) - (1-0.9) \cdot \log\left(\frac{1}{1-0.9}\right) - (1-0.5) \cdot \log\left(\frac{1}{1-0.5}\right) \\ - (1-0.3) \cdot \log\left(\frac{1}{1-0.3}\right)$$

MEASURE OF FUZZYNESS

$$F_F(g) = 1 - \frac{D_F(g, g^c)}{\| \text{SUPPORT}(g) \|}$$

$$D_F(g, g^c) = \left[\sum_{i=1}^n |g(x_i) - g^c(x_i)|^p \right]^{1/p}$$

$$= \sqrt[2]{0.3 - 0.7|^2 + 2|0.5 - 0.5|^2 + 2|0.9 - 0.1|^2 + |1 - 0|^2}$$

=

$$= 1 - \frac{D_F(g, g^c)}{\| \text{SUPPORT}(g) \|} \rightarrow \text{USE EUCLIDEAN DISTANCE THAT IS } \sqrt{\sum_{g(x_i) > 0} (g(x_i) - g^c(x_i))^2}$$

NUMBER OF ELEMENTS IN THE SET

ORDERED WEIGHTED AVERAGE

$$w_1 = w_2 = w_3 = 2/14$$

$$w_4 = w_5 = 1/14$$

$$w_6 = w_7 = 3/14$$

$$1 \cdot \frac{2}{14} + 0.9 \cdot \frac{2}{14} + 0.9 \cdot \frac{2}{14} + 0.5 \cdot \frac{1}{14} + 0.5 \cdot \frac{1}{14} + 0.3 \cdot \frac{3}{14} + 0.3 \cdot \frac{3}{14} = 0.6$$

CENTER OF GRAVITY

$$\sum \frac{(M(x) \cdot x)}{\sum (M(x))} = \frac{0.3 \cdot 5 + 0.5 \cdot 6 + 0.9 \cdot 7 + 1 \cdot 8 + 0.9 \cdot 9 + 0.5 \cdot 10 + 0.3 \cdot 11}{4.4} = 8 \quad (\text{AS EXPECTED})$$

$\uparrow$
CARDINALITY

To DO 3

- 4) Let us consider a population of integer expressions (for instance $(3+4)-(6+8)$) with operations in $\{+, -, *, /\}$ and numbers in $\{0, \dots, 9\}$. Apply simple Genetic Programming with the following parameters
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